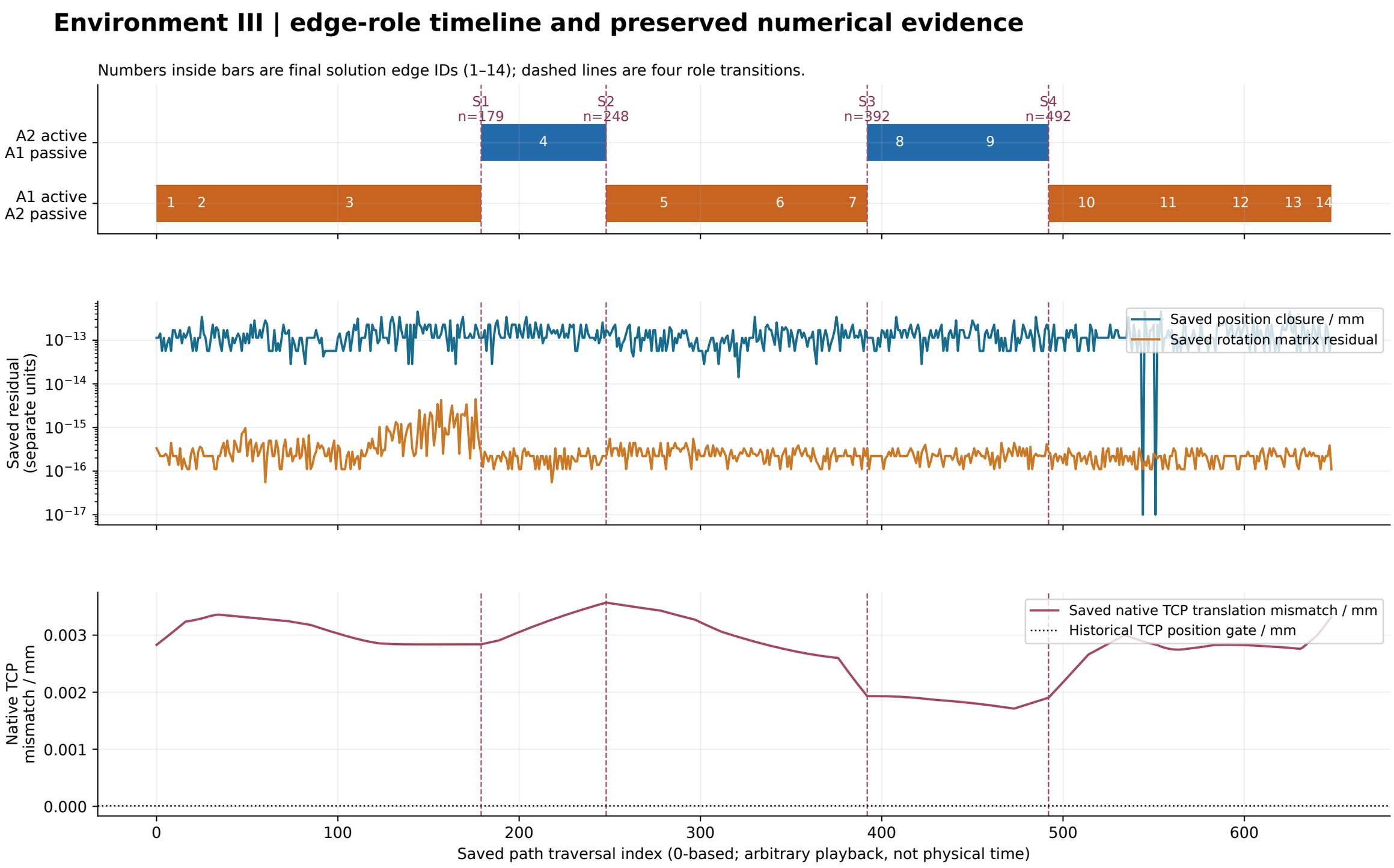
Environment III | edge-role timeline and preserved numerical evidence
Numbers inside bars are final solution edge IDs (1–14); dashed lines are four role transitions.
S1
n=179
S2
n=248
S3
n=392
S4
n=492
A2 active
A1 passive
A1 active
A2 passive
1
2
3
4
5
6
7
8
9
10
11
12
13
14
Saved residual
(separate units)
10^{-13}
10^{-14}
10^{-15}
10^{-16}
10^{-17}
Saved position closure / mm
Saved rotation matrix residual
Native TCP
mismatch / mm
0.003
0.002
0.001
0.000
Saved native TCP translation mismatch / mm
Historical TCP position gate / mm
0
100
200
300
400
500
600
Saved path traversal index (0-based; arbitrary playback, not physical time)
649 aligned pass / 0 native pass; retained independent qualification does not certify continuous CCD, all entity surfaces, or minimum clearance.